

Investor Sentiment and its Role in Asset Pricing: An Empirical Study for India

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Abstract

In this paper, we experiment with alternative investor sentiment indices and their construction methodologies. We further evaluate the role of sentiment based factor in asset pricing for explaining prominent equity market anomalies in India. The findings confirm that our composite sentiment index leads market performance. It also seems to be a better sentiment measure than those currently in vogue in investment literature. Three prominent equity market anomalies namely size, value and price momentum are verified and confirmed for India and the empirical results seem to be robust for alternative portfolio formations. The Fama French 3 factor model performs well in explaining asset returns for different portfolio formations with a few exceptions. We further show that new Fama French 5 factor (2014) model is able to capture returns on some of the unexplained portfolios, while others are explained by our sentiment factor. Our sentiment based factor seems to proxy for the price over-reactions. We thus confirm the role of sentiment factor in asset pricing especially for small stock portfolios. The findings are pertinent for global investors, policymakers as well as academic community. The study contributes to asset pricing and behaviour finance literature for emerging markets like India.

Keywords: investor sentiment, equity pricing anomalies, CAPM, Fama French Model, behaviour finance

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Section 1: Introduction

The tenets of traditional classical finance rest on Efficient Market Hypothesis (Fama 1970) which states that the market price at any time instant reflects all available information in the market. It assumes risk averse rational investors and even if some investors give supply or demand shocks by trading irrationally, the rational arbitragers absorb these shocks to bring the security prices back to its fundamental levels. Heavy literature has been developed on the effectiveness of the classical finance theory to explicate in a meaningful way the commonly cited stylized facts of the stock markets. However, traditional finance theorists failed time and again to explain unconventional market movements leading to the genesis of new field of finance i.e behavior finance.

Behavioral theories predicate that investors may form erroneous stochastic beliefs, either with excessive optimism or pessimism, and therefore incorrectly evaluate asset values, causing asset prices to deviate from their intrinsic values (see, e.g., (De Long et al. ,1990), (Lee et al. ,1991), and (Kumar and Lee, 2006)). In early contributions, sentiment was linked to speculative bubbles (Smidt, 1968), biased expectations (Zweig, 1973), and noise (Black, 1986). As sentiment is related with different attributes, there is no universal definition accepted by literature. In DeLong et al. (1990), it is investors' formation of beliefs about future cash flows and investment risks that are not justified by existing facts. For Shleifer (2000), it 'reflects the common judgment errors made by a substantial number of investors, rather than uncorrelated random mistakes'. In Brown and Cliff (2004), it represents the 'expectations of market participants relative to a norm: a bullish (bearish) investor expects returns to be above (below) average, whatever average may be'. In Baker and Wurgler (2006), it is the propensity of investors to speculate which describes waves of optimism and pessimism. There is an increasing literature on investor sentiments to explain the cross section of stock returns. Overall the results are not uniform as they might be biased towards the approach followed to construct the sentiment index and level of market development in respective countries. However, most of the research on sentiment measurement focuses on the U.S. and other developed stock markets and rely on the notion that it is mainly retail investors who are affected by

sentiment waves and who cause stock prices to drift away from their fundamental values (Kumar and Lee, 2006). They implicitly assume that the institutional investors are informationally more rational in their trading behaviors whereas the sentiment wave is more prevalent amongst the retail investors. Therefore it is important to test the robustness of the findings from the developed markets in the Indian context given the different market microstructure settings and composition of investor population. Specifically, an emerging market like India has some unique characteristics like high market volatility and the percentage of retail participation in stock markets¹ is a mere 2% of its total population compared to other major capital markets (USA, UK, Europe, Japan) where it is more than 20%. This gives rise to a question whether investor sentiments play any role in Indian context.

To empirically investigate the impact of investor sentiment on the Indian stock market, we first construct the investor sentiment index. The measurement of investor sentiment is difficult and the literature has proposed numerous sentiment proxies (Qiu and Welch, 2006). We adopt different approaches followed by Baker and Wurgler (2006), Baker; Wurgler and Yuan (2011), Huang; Jiang; Tu and Zhou (2013) and also construct our own composite sentiment index for India. The following objectives are set out: 1. Evaluate the relationship between the investor sentiment and market performance 2. To check the lead/lag relationship between the various sentiment indices so constructed 3. To test the presence of asset pricing anomalies (size, value and price momentum) in the Indian capital market and if they can be explained by alternative asset pricing models 4. To examine if asset pricing results exhibit partition sensitivity i.e they differ for alternative portfolio formations (5/10/20 portfolios) 5. To examine the role of investor sentiment factor in stock returns.

The rest of the paper is organized as follows. Section 2 covers a brief review of literature. Section 3 outlines the data description and its sources. Section 4 describes the construction of investor sentiment index and its relationship with market performance. In section 5, we assess in detail the role of systematic risk factors as well as sentiment factor in the asset pricing framework. The last section contains summary and concluding observations.

Section 2: Review of Literature

Investor sentiment measures can be classified into three categories i.e direct, indirect and meta measures, broadly by how the sentiment measures are collected. Direct or survey-based sentiment measures include Investor's Intelligence survey and the American Association of Individual Investors survey (Clarke & Statman, 1998; Fisher & Statman, 2000; Lee et. al., 2002; Baker & Stein, 2004; Brown & Cliff, 2004). Two popular survey-based sentiment measures in the market are the ING Investor Dashboard sentiment index² and Investors Intelligence sentiment index³. A number of empirical studies use market-based indicators as an indirect measure of sentiment. Although many of them use various proxies to measure investor sentiment, their common conclusion is that sentiment can be measured and it correlates highly with stock returns. Some researchers (Zweig, 1973; Lee et al, 1991) have used closed end fund discount (CEFD) as sentiment proxy. Researchers (Pan and Poteshman, 2006) have also used derivatives data (put call ratio, VIX, open interest and options premium) as sentiment proxy. Brown and Cliff (2004) used various sentiment proxies to analyze the relationship between investor sentiment and stock returns. These include: survey data from Investor Intelligence; technical indicators such as the ratio of the number of advancing issues to declining issues; variables that relate to particular types of trading activity such as margin borrowing, the percent change in short interest and the ratio of specialist's short sales to total short sales; and many others. Baker and Wurgler (2006, 2007) construct a composite index of sentiment based on the first principal component of six variables namely the closed-end fund discount, NYSE turnover, the number and average first-day returns on IPOs, the equity share in new issues and the dividend premium. Baker et. al. (2011) used 4 proxies (volatility premium, total number of IPO, first day IPO returns, market turnover) to construct investor sentiment indices for 6 major international stock markets. The third category is a combination of both the direct and indirect sentiment proxies in the form of a composite sentiment measure.

Early empirical evidence centered on demonstrating how sentiment predicts future returns in the US stock market (Kothari & Shanken, 1997; Neal & Wheatley, 1998; Shiller, 2000; Brown & Cliff, 2005) and estimating the effect of sentiment on small-stock premiums (Lee et. al., 1991; Swaminathan, 1996; Neal & Wheatley, 1998; Brown & Cliff,

2004). Schmeling (2009) found, for 18 industrialized countries, that investors' sentiment acts, on average, as a significant predictor for stock returns. Zhu (2013) models time-varying return distribution through quantile regressions and copulas and uses sentiment index as a potential candidate to predict the return distribution. Huang et. al. (2013) proposed a new investor sentiment index which had a great predicting power to predict cross sectional stock returns than the existing indices. On the contrary, Elton et al. (1998) conclude that sentiment is subsumed by other risk factors in a well specified model of asset return behavior. Kling and Gao (2008) showed that investors' sentiment does not influence stock returns for China. Finter et. al. (2010) also could not find much predictive power of sentiment for future stock returns in German stock markets. Another set of studies examines the possibility of a causal relationship between index returns and changes in investor sentiment, failing to find any sentiment effect on short-run returns (Brown & Cliff, 2004; Wang et. al. 2006). In the Indian context, Sehgal et al (2009, 2010) have made an early attempt to develop a sentiment index to confirm the bilateral causality between investor sentiment and index returns. Dash and Mahakud (2012) also developed a sentiment index from the market related proxies and confirmed the unidirectional causal relationship between sentiment index and the two benchmark market indices in India. Dash and Mahakud (2013) investigated the impact of investor sentiment on the cross section of stock returns in India and concluded that stocks which are hard to value and difficult to arbitrage with characteristics which are significantly influenced with the sentiment risk. There is no theoretical argument on the number of proxy variables that can be used to construct investor sentiment index and the literature regarding the estimation methodology to extract this sentiment component is also evolving. Given the heavy literature in the developed countries wherein the role of investor sentiment is documented in the asset pricing framework to explain the cross section of returns, it is logical to extend the work in the context of emerging market i.e India where it is only nominally tested. Our study makes the following significant contributions: 1. Innovations in measuring investor sentiment (both in terms of number of sentiment proxy variables that are included and contributions to methodologies proposed in the literature) 2. Besides testing for asset pricing anomalies using Capital Asset Pricing Model (CAPM) and Fama French (FF, 1993) model, recently introduced FF 5 factor

model (FF, 2014) was tested for its ability to explain cross section of asset returns 3. Corporate profitability dimension used by FF 5 factor model is attributed (Walkshausl, 2013) as a measure of firm quality hence we modify the FF 5 factor model by using an alternative measure of firm quality that is cashflow variability as proposed by Walkshausl 4. Asset pricing anomalies were tested for robustness relating to partition sensitivity that is construction of alternative number of portfolios (5/10/20) from the dataset 5. Sentiment factor is added to multifactor models to verify if it plays a role in asset pricing 6. We examine post holding return pattern of sample portfolio(s) whose returns were not explained by the risk models to obtain a better interpretation of our sentiment factor.

Section 3: Data Description and Sources

As there is no universal proxy for measuring investor sentiment hence we construct four alternative measures of investor sentiment i.e BW index (following Baker et. al (2006)), BWY index (following Baker et al. (2011), Aligned Sentiment index (following Huang et. al. (2013) and our Composite sentiment index (using all available market related proxies surveyed from literature). The selected market related proxies (Refer Exhibit 1) are: market turnover (TURN), advances to decline ratio (ADVDEC), put call ratio (PCR), bank deposits to market capitalization (BDEPMCAP), net equity investment in capital markets by mutual funds (EQINVMF), total volume of IPO (IPOVOL) and their 1st day returns (IPO1DRT), equity issue in total issue (EQTOTAL), dividend premium (DIVPREM) and volatility premium (VOLPREM). A total of 150 monthly observations are obtained from various data sources covering the period from July 2001 to December 2013. The starting date is carefully chosen as the trading system in India changed from forward trading in specified shares⁴ to cash trading for all shares with effect from July 2001.

Following Baker and Stein (2003) who suggested that market turnover, or more generally liquidity, can serve as a proxy and is expected to increase when investor sentiment is high. ADVDEC is used as a proxy and a rising (declining) values of the ADVDEC can be used to confirm the upward (downward) trend of the market (Brown and Cliff, 2004). A lower (higher) PCR suggests bullish (bearish) sentiment in the market (Brown and Cliff, 2004; Finter et al., 2010). Also a lower (higher) BDEPMCAP ratio confirms the bullish (bearish) trend in the market.

Consistent with Brown and Cliff (2004) and Finter et al. (2010), EQINVMF is considered as another sentiment proxy for the mutual fund industry and a higher number indicates positive sentiment.

Following Baker and Wurgler (2006, 2007) and Finter et al. (2010), we take the IPOVOL and IPO1DRT as another sentiment proxies. Loughran and Ritter (1995) state that corporate executives time their IPOs to take advantage of fluctuations in investor sentiment. Similarly a higher EQTOTAL ratio can be considered as a bullish market sentiment (Baker and Wurgler, 2006, 2007).

DIVPREM and VOLPREM are constructed in the following manner: In March of year $t-1$, the securities constituting BSE500 are ranked on the basis of the stylised characteristic under consideration (dividend, volatility). The ranked securities are then classified into five portfolios (quintiles) P1 to P5 (P1 being the top quintile and P5 being the lowest) and equally-weighted monthly average market to book ratios are estimated for these portfolios for the next 12 months (t). The portfolios are rebalanced on the 31 March of t year. Baker and Wurgler (2004) use dividend premium variable to proxy for relative investor demand for dividend-paying stocks. We have characterized the same as high and low dividend paying firms as more profitable firms with weaker growth opportunities (Fama and French (2001)), the dividend premium may proxy for the relative demand for this correlated bundle of characteristics. An increase in the ratio indicates increased apprehensions and, therefore, a decrease in investor sentiment. Motivation of the VOLPREM is derived from the theoretical prediction that sentiment has its strongest effects on hard to value and hard to arbitrage stocks. Volatile stocks are inherently riskier to trade—volatility brings with it fundamental and arbitrage risk, as in Pontiff (1996). Volatile stocks also tend to be costlier to trade. Bid-ask spreads are wider due to the probability of informed trading (Glosten and Milgrom, 1985). All else equal, high volatile stocks are those where noise traders can defend extreme values for, as befits their current optimism or pessimism.

We deregress the sentiment proxy variables for macroeconomic information to eliminate business cycle fluctuations. The list of macroeconomic variables used for this purpose, alongwith their description and data sources, is given in Exhibit 2. BSE SENSEX, a

benchmark index is used as a market proxy for market performance as it attracts the FII flows and thus is more influenced by sentiment factor.

For the asset pricing tests, data is employed for 497 companies which form part of BSE500 index. The study uses month end closing adjusted⁵ share prices from July 2001 to Dec 2013. The sample set accounts for more than 90% of market capitalization as well as average daily trading volume on the Indian exchanges and hence is fairly representative of market performance. The month end share price series have been converted into percentage return series for further estimation. The Bombay Stock Exchange (BSE) 500 index, a broad based and value weighted index, is used for asset pricing tests as it acts as a proxy for market portfolio. Data is collected for following company characteristics which are used to form stylized portfolios as well as the risk factors:

1. Market capitalisation (as size proxy) is calculated as the natural log of price times shares outstanding.
2. Price to book value per share (inverse of Book Equity/Market Equity) (as value proxy) represents the security price over a company's book value
3. Average of trailing 6 month's returns (as momentum proxy) is calculated
4. Return on equity (as profitability proxy) is calculated as the income available to common stockholders for the most recent fiscal year divided by the average common equity and is expressed as a percentage.
5. Cash Flow variability (as proxy for firm quality) was calculated as a standard deviation of trailing 5 years cashflow from operations for the various companies
6. Change in Total Assets (as an investment proxy) was calculated as difference of total assets between year t and year t-1

Data on share prices, market index and all company characteristics has been obtained from the CMIE Prowess database. The implicit yields on 91-day treasury bills have been used as risk-free proxy as is the standard practice in finance literature. The data for this has been taken from the RBI monthly handbook of statistics.

Section 4: Investor Sentiment Index- Construction and Relationship with Market Performance

4.1 Measuring Investor Sentiment

As is evident that the various sentiment proxies are measured on different scales hence we first standardize each indicator before it is used in our further analysis. To ensure that our sentiment proxies are not driven by fluctuations in macroeconomic conditions rather than only the non fundamental component of sentiment, we adjust our sentiment proxies for influence of business cycle fluctuations. Following Baker and Wurgler (2006) and Dash and Mahakud (2012), we regress these standardized proxies against the market variables as shown below:

$$\text{Sent}_{i,t} = \alpha + \beta_{1,i} \text{IIP} + \beta_{2,i} \text{FX} + \beta_{3,i} \text{WPI} + \beta_{4,i} \text{M3} + \beta_{5,i} \text{TERM} + \beta_{6,i} \text{EXIMP} + \beta_{7,i} \text{FII} + \beta_{8,i} \text{D} + \varepsilon_{i,t} \quad (1)$$

In this regression, $\text{Sent}_{i,t}$ is one of the many sentiment proxies whereas the explanatory variables are as described above. Consistent with the related literature (Baker and Wurgler, 2006, 2007; Brown and Cliff, 2004), for our subsequent analysis, we will use residuals from the analysis ($\varepsilon_{i,t}$) as the orthogonalised sentiment indicator proxies. We propose to construct 3 alternate variants of sentiment index using Baker and Wurgler (2006) methodology. BW index (following Baker et. al (2006)) uses the following proxies: turnover, number of IPO, IPO first day returns, dividend premium and equity to total issues to construct a sentiment index⁶. BWY index is constructed (following Baker et al. (2011) using the following proxies: turnover, volatility premium, number of IPO and IPO 1st day returns. Composite index is proposed by taking all sentiment proxies discussed in section 3. Summary statistics of the individual sentiment proxies are presented in Table 1. Correlations between all orthogonalised sentiment proxies are presented in Table 2. They show that our sentiment proxies are strongly correlated. Subsequently, we perform principal components analysis (PCA) for measuring the common variations and to isolate the common components in these sentiment proxies. The principal component analysis filters out idiosyncratic noise in the orthogonal sentiment proxies and captures their common component. Also the relative timing of the given orthogonalised proxies may reflect a given shift in the sentiment earlier than others. Following Baker and Wurgler (2006), all the respective proxies and their lags are sent as inputs for the PCA to compute the first stage index containing loadings for the same inferred from the rotated component matrix which shows on which component the variable loads (unlike the Baker and Wurgler (2006) which considers only the first

principal component to compute the factor loadings). Subsequently we compute the correlations between the first stage index and the current and lagged proxies. We then compute the final sentiment index from the respective loadings inferred from the rotated component matrix of the proxies—each respective proxy’s lead or lag, whichever has higher correlation with the first-stage index. Thus the 3 sentiment indices (BW index, BWY index and Composite index) are thus formed.

In Baker and Wurgler (2006), investor sentiment is estimated as the first principle component (PC) of the cross-section of the various proxies that maximally explains the total variation of all the proxies. The principal component can potentially contain a substantial amount of common approximation errors of these proxies which are part of their variations that are not relevant for forecasting returns. Following Huang et. al. (2013), we extract the information in the proxies that is relevant to the expected stock returns from the error or noise. Statistically, the partial least squares (PLS) method pioneered by Wold (1966, 1975) and extended by Kelly and Pruitt (2012, 2013) is used. PLS can be implemented in the following 2 steps of OLS regression. In the first step, for each orthogonalised investor sentiment proxy i (10 variables in our case), we run a time series regression of sentiment proxies ($X_{i,t-1}$) on a constant c and realized index return R_{sensex} i.e BSE SENSEX in our case.

$$X_{i,t-1} = \mu_{i,0} + p_i R_{\text{sensex}} + \varepsilon_{i,t-1} \quad (2) \quad \text{for } i=1 \text{ to } N \text{ where } N \text{ is number of proxies}$$
 This regression is performed as rolling regressions⁷ with a window size of 24 months (2 years) to get vector of factor loadings p for each sentiment proxy i . These loadings capture the sensitivity of each proxy to the investor sentiment driving the future market return. In the second step, for each time period t , we run a cross sectional regression of $X_{i,t}$ on the corresponding loading p_i estimated in the first stage regression. Thus the first stage loadings becomes an independent variable and the investor sentiment index (aligned investor index) is the coefficient to be estimated i.e . H_t

$$X_{i,t} = c_t + H_t \hat{p}_i + v_{i,t} \quad (3) \quad \text{for } t=1,2,\dots,T$$

Thus we now have constructed all the four variants of sentiment index (BW Index, BWY index and Composite Index based on PCA procedure and Aligned Sentiment index based on PLS procedure)

The factor loadings of the 3 indices constructed on the basis of PCA are shown in Table 3. One can clearly see that the composite index explains maximum (67%) of the sample variance amongst the other indices. Also it can be seen that correlations between the first stage index and subsequent final index are highest for composite index (0.90) and thus we are not running risk of losing maximum information while factoring in the relative timing of the variables. The composite aligned investor sentiment index is also computed based on the PLS methodology.

4.2 Investor Sentiment Indices and Market Performance

Literature suggests a long run equilibrium relationship between investor sentiment and market returns. We empirically evaluate the relationship between the various variants of the investor sentiment so constructed and the stock market proxied by BSE SENSEX. To examine this relationship, we employ the autoregressive distributed lag model (ARDL) suggested by Pesaran et al. (2001) for cointegration investigation and error correction (short run) analysis. One advantage of this technique is that it doesn't consider the problems arising from the different order of integration of the variables. We use the variables in natural logarithm form to assess the significance of the relationship between the investor sentiment indices and the benchmark index return. The concept of co-integration becomes relevant when the time series being analyzed are non stationary. The testing for stationarity of the data was done through the Augmented Dickey Fuller (ADF) test. To investigate the presence of long-run relationship, bound testing under Pesaran, et al. (2001) procedure is used. An ARDL representation of the relationship is formulated as follows:

$$\Delta \text{Sensex}_p_t = \alpha_0 + \sum_{i=0}^n \alpha_{1i} \Delta \text{Sensex}_p_{t-i} + \sum_{i=0}^n \alpha_{2i} \Delta \text{SentimentIndex}_{t-i} + \beta_1 \text{Sensex}_p_{t-1} + \beta_2 \text{SentimentIndex}_{t-1} + \varepsilon_t \quad (4)$$

where Δ denotes the first difference operator, α_0 is the drift component and ε_t is the usual white noise residuals. The lag length of the ARDL model is so selected where Schwarz Information Criteria (SIC) is minimized.

The ECM version of modified ARDL is used to investigate the short run dynamic relationships. All this will be done through the ECM (Error Correction Mechanism) applied through the Ordinary Least Square (OLS) method. The error correction version of ARDL model explained above is as follows:

$$\Delta \text{Sensex_p}_t = \alpha_0 + \sum_{i=0}^n \alpha_{1i} \Delta \text{Sensex_p}_{t-i} + \sum_{i=0}^n \alpha_{2i} \Delta \text{SentimentIndex}_{t-i} + \lambda \text{EC}_{t-1} + \mu_t \quad (5)$$

where λ is the speed of adjustment parameter and EC is the residuals that are obtained from the estimated long run equilibrium model (equation 4) explained above. The coefficient of EC term acts as evidence of direction of casual relation and reveals the speed at which discrepancy from equilibrium is corrected or minimized. The direction of causality is confirmed from the Granger Causality test between the various variants of sentiment index and the benchmark stock market index.

ADF tests (refer table 4) confirmed that the time series of the four variant of sentiment index so constructed were all stationary at level and thus integrated to order 0. The market index time series was however found to be stationary at first difference and hence integrated to order 1. For this reason, ARDL approach is used for the test of cointegration and results of table 5 (Panel A) confirms the presence of long run equilibrium relationship between 4 variants of the investor sentiment index individually with the SENSEX prices as the F statistic is greater than the critical values. Panel B of table 5 results show for short run dynamics of these cointegrated series that except for composite index (for which absolute value of ECT term is smaller than that the market index), all the other 3 investor sentiment indices lags the SENSEX prices. Granger causality results (Table 5, panel C) confirm the short run results that the causal relationship is from the composite index to the SENSEX returns. Hence composite index is the only sentiment index which leads the SENSEX prices and can help predict the SENSEX returns.

To test the robustness of our results that composite index outperforms other investor sentiment indices so constructed, we evaluate the long run equilibrium relationship and short run dynamics between our composite index and other 3 sentiment indices so constructed. Results (Table 6, Panel A) indicate that there is a longrun equilibrium relationship between composite index and other sentiment indices. Results of the short run dynamics (Table 6, Panel B) confirm that composite index leads all other 3 indices and hence is a good metric for measuring investor sentiment in India.

In conclusion, our composite index (based on PCA procedure) seems to be a better investor sentiment proxy compared to BW, BWY and Aligned Investor Sentiment index. Further, it leads the market index and hence can be used to predict the market

performance unlike the other sentiment indices which empirically lag compared to the market returns.

Section 5: Role of Risk Factors and Investor Sentiment in Asset Pricing

5.1. Stylised Portfolios

We construct three set of stylized portfolios based on 1. company size (market capitalization) 2. value (P/B) and 3. price momentum (past 6 month average returns). While the size and value portfolios are rebalanced on annual basis, the momentum based portfolios are realigned on 6 monthly basis. The size and value portfolios are formed as follows: In June⁸ of year t (March of year t-1), the securities are ranked on the basis of market capitalization/ PB ratio. The ranked securities are then classified into five portfolios (quintiles) P1 to P5 (P1 being the top quintile and P5 the lowest) and equally-weighted monthly excess returns are estimated for these portfolios for the next 12 months starting from July⁹ of year t. P1 and P5 are referred henceforth as corner portfolios in the study and represent top 5% and bottom 5% stocks on the basis of market cap and P/B ratio. For momentum portfolios, in June of year t, we rank the sample stocks in ascending order on the basis of their average returns during last 6 months and form quintiles (P1 the highest quintile called the winner and P5 the lowest termed the loser). Equally-weighted monthly excess returns are estimated for these portfolios for the next 6 months. The portfolios are again rebalanced at December of year t and winners and losers are so formed. Thus a 6/6 strategy is followed where formation and holding period are kept as 6 months each.

Asset pricing anomalies (namely size, value and price momentum) were tested for robustness checks for their partition sensitivity owing to alternate portfolio formations. The securities were now ranked on the basis of stylized characteristics (size, value and momentum) as discussed above and were now classified into 10 portfolios (deciles) P1 to P10 and 20 portfolios (vigintiles) P1 to P20. P1 is the portfolio consisting of top 10/5% (decile/vigintile) of companies with highest attribute while P10/ P20 (decile/vigintile) consist of lowest 10/5 % companies with lowest attribute under consideration.

The unadjusted average returns on size, value and momentum for alternative portfolio formations (quintiles, deciles, vigintiles) are shown in table 7. The mean unadjusted monthly returns on size sorted portfolios for quintiles are monotonically increasing from

large stocks (1.33%) to small stocks (4.09%) thereby confirming the negative relationship between size and average returns. The monthly return differential between small and large stocks is 2.76% thus confirming the size effect as postulated by Banz (1981) and confirmed by Roll (1981), Lettau and Ludvigson (2001), Sehgal and Tripathy (2005). Unadjusted average returns for quintiles are also monotonically increasing from 1.86% per month for high P/B stocks (low Book to market equity) to 3.03% per month for low P/B stocks (high BE/ME) thus generating a return differential of 1.17% a month. This confirms the presence of strong value effect as postulated by Stattman (1980) and confirmed by DeBondt and Thaler (1987), Lakonishok et al. (1994). But clearly size effect dominates the value effect by 2.36 times which is in confirmation with the findings for the emerging markets including India (refer Sehgal et. al (2012)). Unadjusted returns on momentum sorted portfolios show that the monthly mean returns for the winners portfolio (P1) is 2.73% per month whereas the monthly mean returns for the losers portfolio (P5) is 1.37% per month thus providing a momentum profit of 1.36%. Thus the size, value and momentum effects are empirically confirmed for the Indian markets for the quintile portfolio.

Panel B and C of table 7 provides the unadjusted average returns on characteristics based portfolios for deciles and vigintiles formations respectively. The unadjusted return differential between small and large stocks (removing P1 from decile and P1, P2 from vigintile) is 3.07%/3.42% monthly for both decile/vigintile portfolios confirming the negative relationship between size and average returns. Observing raw returns for deciles, P1 comprises of large firm stocks chased by institutional investors. Excess demand for them leads to positive price reaction resulting in higher expected returns. This leads to increased trading activity and the extra returns may be a mere compensation for increased volatility. Results are reconfirmed for vigintile portfolio where P1 and P2 portfolios are equivalent to decile P1 portfolio (in terms of number of constituent stocks). Thus, size effect is almost monotonic from P2 to P10 in deciles and P3 to P20 in case of vigintiles. The monthly unadjusted return differential between low P/B and high P/B stocks (removing P1 from decile and P1, P2 from vigintile) is 1.65%/2.31% for decile/vigintile portfolios thus confirming the value effect. The size effect is thus 1.86 times the value effect for the decile portfolios and 1.48 times for the vigintile portfolio. Again in case of

P/B ranked stocks, P1 (high P/B stocks) are actually growth stocks which catch institutional investors fancy. Higher demand leads to positive price reactions and higher returns but if higher trading activity results in greater price volatility, the extra observed returns may be mere compensation for extra risk. Removing P1, the relationship is almost monotonic from P2 to P10 for deciles. Similarly removing P1 and P2 (equivalent to decile portfolio P1), returns increase almost monotonically from P3 to P20. Monthly unadjusted return differential between winners and losers portfolios (removing P1 from decile and P1, P2 from vigintile) is 1.73%/3.22% for decile/vigintile portfolios confirming the momentum effect. Observing raw returns, P1 (winners) have moderate expected returns because probably investors do not believe that they can sustain the higher returns hence price correction process will compel their returns to sober down. The momentum relationship is almost monotonic from P2 to P10 in case of deciles. Similarly removing P1 and P2 (equivalent to decile portfolio P1), returns decline almost monotonically from P3 to P20. In sum, the size, value and momentum premiums are strongest for vigintile portfolios thus confirming that performance of characteristics sorted portfolio is sensitive to alternate portfolio constructions.

5.2 Tests of CAPM

CAPM regressions are run on the sample portfolios using the “excess return” version of the market model.

$$R_{pt} - R_{ft} = \alpha + \beta (R_{mt} - R_{ft}) + \varepsilon_t \quad (6)$$

Where $R_{pt} - R_{ft}$ denotes monthly excess returns on the portfolio over risk free return (R_{ft}) and $R_{mt} - R_{ft}$ denotes monthly excess market return. CAPM postulates that excess returns on the portfolio should be fully factored in the by the excess market returns leading α to be zero. Any significant positive α would lead to extra normal profits pointing towards CAPM anomaly.

The CAPM results for all alternate portfolio formations are presented in Table 8. It can be seen that even after adjusting for market risk, small size stocks on an average earn monthly 2.8% return (which is statistically significant) compared to 0.15% for big stocks in case of quintiles. Adjusted R square is low for small stock quintile portfolio (P5) vis-a-vis large stock portfolio (P1) showing that the former has a very large unexplained variance in their returns. CAPM results show that intercept value is low for high P/B

portfolios than low P/B portfolio for quintiles indicating the latter is generating higher market risk adjusted returns than the former. However, CAPM is unable to absorb the cross sectional return differences on any of the value sorted portfolios for quintiles. CAPM results for momentum sorted portfolios for quintiles show that intercepts of the winner portfolio are statistically significant thus confirming that market factor does not explain momentum. This shows the presence of strong momentum profits as postulated by Jegadeesh and Titman (1993) and confirmed by Chordia and Shivkumar (2002)

Panel B and Panel C of table 8 discusses the CAPM results of decile and vigintile portfolios respectively. CAPM results show that even after adjusting for market risk, small stocks portfolio in deciles/ vigintiles (P10/P20) on an average earn higher returns (3.42%/3.63%) compared to large stocks i.e P1 (1.64%/1.34%). The small stock portfolios earn statistically significant positive extra risk adjusted returns in both the cases (decile and vigintile) thereby confirming the size effect. The results convey that intercept value is low for high P/B (P1) portfolios than low P/B (P10/P20) portfolios indicating the latter is generating higher market risk adjusted returns than the former in both deciles/vigintile portfolios. The intercepts of the winner (P1) and loser (P10/P20) for both the decile and vigintile portfolios become insignificant in the CAPM framework and thus it helps to capture the momentum effect. In sum, CAPM is able to explain only 42 out of the total 105 portfolios (5/10/20 portfolios based on size, value and momentum characteristics respectively) however it is unable to absorb cross sectional differences on all value sorted portfolios for almost all the three alternate portfolio formations.

5.3 Tests of Fama French 3 Factor Model

Subsequently, we proceed to make use of Fama and French (FF, 1993) 3 factor model (FF3f) to check whether it can capture the returns which are missed by one factor CAPM. The FF3f model equation is

$$R_{pt} - R_{ft} = \alpha + \beta (R_{mt} - R_{ft}) + \gamma SMB_t + \delta LMH_t + \varepsilon_t \quad (7)$$

Where SMB_t and LMH_t are the monthly return on the size and value (P/B) mimicking portfolios. We use LMH factor instead of HML factor in the FF3f regression. Hence our interpretation of the value factor will be inverse. Following Sehgal et. al. (2012), we construct a 2*2 size-value partition. SMB and HML are estimated as follows. In each year of the sample period t , the stocks are split into two groups- big (B) and small (S) -

based on whether their market capitalization at the end of June of every year in the sample period is above or below the median for the stocks of the companies included. The price to book equity ratio is calculated in this month for all the companies. The stocks are now split into two equal P/B groups low (L) and high (H). Then we construct four portfolios viz. S/L, S/H, B/L, B/H from the intersection of the two size and two P/B groups. Monthly equally weighted return series are calculated for all portfolios from July of year t to June of year t. The simple average of the monthly returns of the two big size portfolios (B/L, B/H) is subtracted from the average of the two small size portfolios (S/L, S/H) to get the monthly return of the SMB factor. This factor is free from value effects as it has about the same weighted-average price to book.

$$\text{SMB} = (\text{S/L} + \text{S/H})/2 - (\text{B/L} + \text{B/H})/2$$

Similarly we get the LMH factor which is independent of the size factor.

$$\text{LMH} = (\text{S/L} + \text{B/L})/2 - (\text{S/H} + \text{B/H})/2$$

The SMB and LMH factors exhibit a correlation of -0.14 and hence are non overlapping by construction.

A significant intercept term α in the CAPM regressions which becomes insignificant in the FF framework signals that the unexplained cross section patterns of the average stock returns are priced in the FF model. A significant intercept term in FF model on the other hand ascertains the need for an enhanced factor structure. Greater sensitivity of the portfolio returns are shown by the higher factor loading in the size and value factors.

FF3f model results for alternate portfolio formations are presented in Table 9. FF3f model only partially explains the size effect in case of quintiles, as the small stock portfolio (P5) still provides an extra normal return of 0.59% per month (t stat 2.43) which is statistically significant. The loadings for SMB and LMH factors on the low P/B portfolios (P5) are more than double than those for high P/B portfolio (P1) for the quintiles thereby confirming the role of size and value factors in returns. The FF3f model explains the cross sectional difference in returns for all value sorted quintile portfolio and the intercept coefficients become insignificant. The statistically significant intercepts of the momentum sorted portfolio returns also become insignificant in the FF3f framework and thus the model explains the momentum effect for quintiles.

Panel B and Panel C of table 9 discuss the FF3f model results of decile and vigintile portfolios respectively. The cross section of average returns for various size sorted portfolios for deciles and vigintiles which were missed by CAPM were absorbed by FF3f model except for P9 portfolio in case of deciles and P4, P18 and P19 in case of vigintiles. FF3f results show that SMB coefficient is higher for P10/P20 as compared to P1 (2.29/3.55 times coefficient of P1), confirming role of size factor in explaining small firm effect. Contrary to expectations, the big firm stocks (P1) seem to load on LMH factor as shown by LMH loadings in case of vigintiles. Both the size and value factors contribute towards explaining the earlier statistically significant extra normal returns which die out as we move to FF3f framework for value sorted decile and vigintile portfolios (except for P18). Thus FF3f model captures fully/partially the value effect in case of deciles/vigintiles. The cross section of average returns of the momentum sorted portfolios missed by CAPM was successfully captured by FF3f model.

The size factor seems to dominate the inverse value effect in our case contributing to the overall success of FF3f model. On risk adjusted basis for size sorted portfolio, if excess demand for large size stocks (P1) results in higher volatility owing to more active trading activity, the extra returns may be mere compensation for the extra volatility risk. In the FF3f framework, extra returns on these stocks seem to be explained by FF size (and sometimes value factor(s)). The size factor of FF3f model seems to capture the additional volatility risk. Hence, P1 stocks in deciles/vigintiles seem to be behaving like small firm stocks as shown by their higher loadings for SMB factor. In the FF framework for value sorted portfolios, the high P/B stocks (P1) which catch institutional frenzy leading to greater price volatility, the size factor seems to capture this excess volatility risk and hence explains their alphas. Thus high P/B stocks (P1 in case of deciles/vigintiles) may be either small firms or seem to be behaving like small firms with the high growth potential.

In sum, the FF3f model is much better descriptor of asset returns vis-à-vis CAPM (explaining the extranormal returns of 57 of the 63 portfolios whose returns were missed by CAPM). It explains the equity market anomalies for all portfolios with the exception of small size stock portfolios (P5 in quintiles, P9 in deciles and P4, P18, P19 in vigintiles) and P18 in the value sorted portfolio. The failure of FF3f model in fully explaining their

returns may warrant a need for an enhanced factor structure which we explore in the next section.

5.4 Tests of Augmented FF Models

Next to mop up the extra normal returns in the portfolios which were missed by FF3f model, we augment the FF 3 factor model by adding momentum factor (MOM) and thus construct the Carhart (1997) four factor model whose equation is:

$$R_{pt} - R_{ft} = \alpha + \beta (R_{mt} - R_{ft}) + \gamma SMB_t + \delta LMH_t + \lambda MOM_t + \varepsilon_t \quad (8)$$

Where MOM_t are monthly excess returns of the winners minus loser based on quintile portfolio formation.

Another extension of the FF model is the newly introduced Fama French (2014) 5 factor model (FF5f) wherein investment and profitability have been added as two additional factors to the FF 3 factor model as shown below:

$$R_{pt} - R_{ft} = \alpha + \beta (R_{mt} - R_{ft}) + \gamma SMB_t + \delta LMH_t + \nu LMH_t^{invest} + \phi LMH_t^{profit} + \varepsilon_t \quad (9)$$

Where LMH_t^{invest} and LMH_t^{profit} are monthly excess returns of low minus high portfolios ranked on investments (proxied by change in total assets) and profitability (proxied by return on equity). Walkshausl uses a firm quality factor to augment FF 3 factor model framework. He employs 2 measures of firm quality¹⁰ profitability (ROE) and cash flow variability. Consistent with (Walkshausl, 2013) framework, we replace the profitability by cashflow variability factor and term new factor structure as modified FF5f model as shown below.

$$R_{pt} - R_{ft} = \alpha + \beta (R_{mt} - R_{ft}) + \gamma SMB_t + \delta LMH_t + \nu LMH_t^{invest} + \eta LMH_t^{cfv} + \varepsilon_t \quad (10)$$

Where LMH_t^{cfv} are monthly excess returns of low minus high portfolios ranked on cashflow variability (proxied by standard deviation of trailing 5 years cashflow from operations)

Table 10 provides the results for our augmented FF3f models. Carhart model and modified FF5f model are not able to account for the cross section of returns for the sample portfolios under consideration which can be seen from their intercept coefficient which are still statistically significant. The cross section of returns of the small stock P9/P18 portfolios in case of decile/vigintile was absorbed by FF5f model and the loadings of investment and profitability factors are significant (though it is negative for profitability factor¹¹). Also the significant intercept term of the size sorted vigintile P4

portfolio dies out in the FF5f framework at 5% level of significance. It can be clearly seen that the extra normal returns for the small stock portfolio (P5 in quintiles and P19 in vigintile) and the low P/B P18 portfolio in vigintile remain unexplained even by modified FF5f model. Next we examine whether our composite sentiment index can explain the sticky intercept for these three remaining portfolios thus implying the role for behavioral factor in stock returns.

5.5 Role of Sentiment Factor in Asset Pricing

We augment FF3f/FF5f model with the investment sentiment factor. Sentiment factor is represented by a dummy variable (dSentiment) which takes a value equal to 1 when our composite sentiment index is greater than its mean value and value equal to 0 otherwise. The mix model comprising of FF fundamental factors (FF3f and FF5f) and our investment sentiment factor is:

$$R_{pt} - R_{ft} = \alpha + \beta (R_{mt} - R_{ft}) + \gamma SMB_t + \delta LMH_t + dSentiment + \varepsilon_t \quad (11)$$

$$R_{pt} - R_{ft} = \alpha + \beta (R_{mt} - R_{ft}) + \gamma SMB_t + \delta LMH_t + \nu LMH_t^{invest} + \phi LMH_t^{profit} + dSentiment + \varepsilon_t \quad (12)$$

Where dummy represents the investment sentiment factor

The results shown in Table 10, Panel D1 and D2 are encouraging. The intercept on small size stock (P5 of quintile and P19 of vigintile) portfolio becomes statistically insignificant thus confirming that the FF5f model performs better when augmented by our sentiment factor. The t statistic of the sentiment factor is 1.29/1.67 (for P5/P19 portfolios), which is still significant at 20%/10% level (2 tail basis) and hence their economic significance cannot be ignored in a multivariate framework. The return on low P/B (P18) portfolio seems to be explained by the investor sentiment augmented FF3f model. The alphas fades out but surprisingly the coefficient of investor sentiment factor is not statistically significant.

We next analyze the post holding patterns in raw returns for the 3 sample portfolios under consideration to achieve a better understanding of the role of investor sentiment factor in returns. For this purpose, we study the post holding patterns in unadjusted returns for size sorted P5/P19 of quintile/vigintile and value sorted P18 of vigintile portfolio. The post holding period is defined to be 24 months assuming 12 months of portfolio holding. Thus after each portfolio rebalance date, we estimate average returns for 36 months where the

first 12 months are classified as holding period followed by months 13 to 36 which qualify as post holding period. The average returns for months 1 to 36 are cumulated. The cumulated returns are then divided by the number of months to calculate the average of cumulative returns which are charted in Figure 1, 2 and 3 for the three portfolios under consideration.

The post holding pattern of the 2 size based portfolio shows short term overreaction upto 10 months followed by a correction in the post holding periods. These mild overreactions both from the upside and the downside are missed by the FF risk factors incase of small stock quintile/vigintile portfolios. These extranormal returns reported by our small stock portfolios are captured by the investment sentiment factor within the multi factor asset pricing framework. As for the value based P18 vigintile portfolio, the price overreaction is similar to size based portfolios but continuous correction of overpricing in the post holding period (as long term overreactions persistent in the post holding period and not captured by our investor sentiment factor) is puzzling and warrants further investigation and could be a possible reason for the investor sentiment factor not being priced. Thus our investor sentiment factor is able to capture price overreactions which are subsequently corrected in the short run (upto 10 months for size based portfolios). In contrast, it is not able to account for price overreactions which are corrected over a long time period in case of value sorted portfolio. Hence there is a scope for improving the investor sentiment factor so that it can capture almost all the behavioral aberrations. Our investor sentiment factor is able to capture price overreactions thus confirming it's a good proxy for the behavioural factor. We confirm the role of investor sentiment/ behavioral factor in stock returns in India.

In sum, security returns in India seem to be predominantly driven by rational factors as confirmed by the empirical success of FF3f model in explaining most of the issues relating to prominent equity market anomalies. This does not seem surprising as Indian market is predominantly dominated by institutional investors who mainly rely on fundamentals. Value and momentum anomaly do not pose serious challenge to FF3f model except low P/B P18 vigintile portfolio. Size anomaly is observed even after using FF3f model and therefore seems to be partly risk driven and partly a behavioural outcome. The former is captured by FF5f model factors ie investment and profitability while the

latter owing to short term overreaction is captured by our sentiment factor. Hence our results reconfirm the quality of our investor sentiment factor and highlight its role in asset pricing framework atleast when dealing with size anomaly.

Section 6: Summary and Concluding Observations

The impact of investor sentiment on stock market price formation has been a subject of interest for both academicians and industry practitioners. Following the genesis of behaviour finance literature, theorists have come out with rich empirical work to confirm possible linkage between aggregate investor sentiment and stock returns in the developed markets. We set out to examine this relationship in the emerging market context i.e India where retail participation in stock markets is low compared to developed economies. In this paper, we use various indirect sentiment proxies surveyed from literature to measure investor sentiment. Monthly data for all these investor sentiment proxies have been taken since July 2001 to December 2013 to construct the 4 alternative variants of investor sentiment index using PCA and PLS methodology. These investor sentiment indices were found to be in long run equilibrium relationship with market performance and the error correction version of the ARDL model shows that only our composite sentiment index leads the market confirming its predictive power. Further our composite sentiment index also leads other sentiment indices. These results are reconfirmed by granger's causality test.

Further in line with the existing literature, the presence of prominent asset pricing anomalies that is size, value and momentum effects were confirmed for Indian capital market. Based on above mentioned stylized characteristics, the securities were ranked and classified into quintiles, deciles and vigintiles. Behaviour of raw returns on the characteristic sorted portfolios show that size, value and momentum effects were robust for alternate portfolio construction procedures. On the risk adjusted basis, FF3f model seems to be a better descriptor of asset pricing compared to CAPM as it explains returns on most of the characteristic sorted portfolios (99 out of 105 sample portfolios). For small size stocks portfolio, unexplained by FF3f model, the alphas are captured by additional risk factors of FF5f model or further augmenting it by sentiment factor. However, for small P/B stocks portfolio, augmenting FF3f model itself by sentiment factor helps to capture the extranormal returns missed by FF3f model. Looking at the post

holding pattern for the 3 portfolios that are not explained even by augmented FF3f models, we observe that price overreactions which are corrected in the short run as in the case of 2 size portfolios are captured by our sentiment factor. As against this, price overreactions which involve a long term correction as in case of the value sorted portfolio pose an empirical challenge. Thus further research on construction of a more comprehensive investor sentiment factor is recommended.

The findings are pertinent for regulators as empirical results confirm that Indian stock market is fairly informationally efficient perhaps due to higher levels of institutional trading. Small firm stocks have information asymmetry thereby creating behavioral aberrations. It is for the regulators to strengthen the corporate governance and make good the corporate disclosures of these small firm stocks to help reduce information asymmetry thereby correcting behavioral aberrations. Small size based portfolios are fertile ground for portfolio managers as they provide trading strategy to make profits out of the behavioral corrections in the market. Momentum and value effects are fairly noise which can be arbitrated away by the managers. For the industry practitioners, Fama French 3 factor asset pricing model seems to be an appropriate performance benchmark. Further prominent equity market anomalies do not seem to provide significant risk adjusted extranormal returns in the multifactor asset pricing framework. From academic perspective, the study is relevant as it finds out that size is more complex anomaly which warrants an expanded factor structure model whereas only a parsimonious asset pricing model is needed for value and momentum effects. It also deals with issues relating to construction methodology of investor sentiment index and its possible role in predicting market returns and pricing financial assets. The study contributes to both asset pricing and behavior finance literature for India, which is an emerging market.

End Notes

1. Report of PWC and ASSOCHAM (2011) on “Indian Capital Markets- Key to Double Digit Growth”. Also refer interview of Mr Ashishkumar Chauhan, CEO and MD, BSE at www.valueresearchonline.com/story/h2_story
2. The ING Investor Dashboard sentiment index is published quarterly, covering 13 markets across Asia, including Australia, China, Hong Kong, India, Indonesia, Japan, Korea, Malaysia, New Zealand, Philippines, Singapore, Taiwan and Thailand. It is based on interviews of mass affluent respondents across all markets.

3. The Investors Intelligence sentiment index is released daily, summarizing the forecasts of a number of newsletter writers who issue independent advice from their publications and commentary. It is based on contrarian propositions, which assumes that a consensus trend is always about to reverse.
4. Prior to July 2001, shares in India were classified into specified and non specified group. Forward trading were allowed only in specified group while non specified group were traded on cash basis. Since July 2001 this distinction has been dissolved and now all shares in India are traded on cash basis in the spot segment of the market which is supported by derivatives trading in the F&O segment.
5. The study uses share prices data which is adjusted for capitalization changes such as dividends, stock splits and rights issue.
6. Baker and Wurgler (2006) used data for closed-end fund discount, which is the average difference between the net asset values (NAV) of closed-end stock fund shares and their market prices. The data for the same could not be found in Indian context hence the variable was dropped.
7. Huang et. al. (2013) used fixed estimate of p_i for estimation of aligned sentiment index. We used a rolling window to obtain time varying estimates of p_i in the aligned sentiment index construction.
8. Following the legal requirement in India, we define a financial year from April t-1 to March of year t
9. For portfolio formation, 3 month gap has been kept between portfolio formation and holding period incase of P/B ratios. This is owing to the fact that P/B ratio is an accounting measure and there is normally a delay in submission of financial statements from the financial closing date i.e March 31st. The time gap is justified so that the investors can obtain the requisite information for portfolio formation purpose.
10. Walkshow 2013 confirms the role of firm quality factor in explaining returns on volatility sorted portfolio. His results are consistent for both measures of firm quality namely profitability and cashflow variability.
11. This is in conformity with the results of Sehgal and Subramaniam (2012) who empirically confirmed a negative relation between profitability and returns in India which was in contrast to prior research for mature markets

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Exhibit 1: Investment Sentiment Proxy Variables

Proxy Variables	Description	Source	Included in Index
TURN	trading volume in INR	Bloomberg	BW,BWY, Aligned,Composite
ADVDEC	the ratio of the number of advancing and declining prices of scrips	NSE website	Aligned,Composite
PCR	the trading volumes of put options to call options	NSE website	Aligned,Composite
BDEPMCAP	ratio of bank deposits to market capitalization	Bloomberg	Aligned,Composite
EQINVMF	the net equity investment in capital markets by mutual funds	SEBI website	Aligned,Composite
IPOVOL	number of IPOs in a given month	AceEquity Database	BW,BWY, Aligned,Composite
IPO1DRT	difference between the returns based on IPO offer price and the initial price of the stock at the beginning of the first trading day	AceEquity Database	BW,BWY, Aligned,Composite

EQTOTAL	the ratio of aggregate equity issuance to total debt	SEBI website	BW, Aligned, Composite
DIVPREM	ratio of the average market-to-book ratios of top quintile and bottom quintile of the BSE 500 stocks when they are ranked on dividend payout ratio on 31 march of t-1 year respectively	CMIE Prowess Database	BW, Aligned, Composite
VOLPREM	ratio of the average market-to-book ratios of top quintile and bottom quintile of the BSE 500 stocks when they are ranked on volatility (standard deviation of trailing 12 monthly returns) ratio on 31 march of t-1 year respectively	CMIE Prowess Database	BWY, Aligned, Composite

Exhibit 2: List of Macroeconomic Variables

Market Variables	Description	Source
IIP	Index of Industrial Production for India	CSO
FX	Exchange Rate USDINR	RBI
WPI	Wholesale Price Inflation	Bloomberg
M3	A measure of money supply that includes M2 as well as large time deposits, institutional money market funds, short-term repurchase agreements and other larger liquid assets	Bloomberg
TERM	Term spread which is the difference between 10 year Government Securities yield and 3 month treasury bill	RBI
EXIM	Total exports to total imports ratio	Bloomberg
FII	Net foreign institutional flow	Bloomberg
D	Dummy for recession (Economic crises 2007-2009)	www.federalreservehistory.org

Table 1: Descriptive statistics of sentiment proxy variables

Sentiment Proxies	Mean	Median	Max	Min	STDEV	Obs
TURN	1.36E+11	1.29E+11	3.58E+11	2.13E+10	7.41E+10	150
ADVDEC	0.933133	0.91	1.8	0.55	0.21874	150
PCR	0.913327	0.968938	1.506198	0.331269	0.220312	150
BDEPMCAP	0.00556	0.004929	0.011906	0.002264	0.00233	150
EQINVMF	-1.60E+09	-2.24E+09	7.89E+10	-7.24E+10	2.02E+10	150
IPOVOL	3.313333	2	18	0	3.437138	150
IPO1DRT	21.46379	17.45513	159.5714	-100	40.64353	118
EQTOTAL	0.740847	1	1	0	0.364676	150
DIVPREM	0.850156	0.766086	2.314041	0.063627	0.475111	150
VOLPREM	1.523433	1.037988	9.861702	0.419813	1.714553	150

Note: Ten investor sentiment variables have been selected for the study based on past research.

Table 2: Correlations between orthogonalised sentiment proxy variables

	ADVDEC	BDEPMCAP	DIVPREM	EQTOTAL	IPO1DRT	IPOVOL	EQINVMF	PCR	TURN
ADVDEC	1								
BDEPMCAP	-0.161**	1							
DIVPREM	-0.143	0.271*	1						
EQTOTAL	0.060	-0.296*	-0.226*	1					
IPO1DRT	0.085	-0.208*	-0.144	0.108	1				
IPOVOL	0.163**	-0.392*	-0.084	0.184*	-0.043	1			
EQINVMF	-0.038	-0.236*	0.013	0.076	0.087	0.030	1		
PCR	0.202*	-0.353*	-0.205*	0.152**	0.299*	0.199*	0.021	1	
TURN	0.069	-0.501*	-0.085	0.236*	0.077	0.183*	0.109	0.271*	1
VOLPREM	-0.019	-0.046	-0.661*	0.138	0.075	0.003	-0.039	0.162**	0.048

Note: * indicates the level of significance at 5%, ** indicates the level of significance at 10%

Table 3: Characteristics of sentiment indices constructed on PCA methodology

Factor Loadings	BW Index		BWY Index		Composite Index	
	DIVPREM	0.503	IPO1DRT(-1)	0.5714	ADVDEC	0.6870
	EQTOTAL(-1)	0.556	IPOVOL	0.3693	BDEPMCAP	-0.1176
	IPO1DRT	-0.536	TURN(-1)	0.4634	DIVPREM	-0.2812
	IPOVOL	0.323	VOLPREM(-1)	0.5227	EQTOTAL(-1)	0.5103
	TURN(-1)	0.413			IPO1DRT	0.5722
					IPOVOL(-1)	0.5381
Explained Variance					EQINVMF	0.5163
					PCR(-1)	0.3271
					TURN(-1)	0.1297
					VOLPREM	0.3055
Correlation with respective first stage index						
	64.16%		58.08%		67.39%	
	0.82		0.88		0.90	

Table 4: Results of unit root tests for sentiment indices and BSE SENSEX

	Level t- statistics	First Difference t- statistics	Inference on Integration
BW Index	-5.7465*	-	0
BWY Index	-5.3687*	-	0
Composite Index	-7.0996*	-	0
Aligned Investor Index	-9.3594*	-	0
Sensex	-2.4888	-12.0489*	1

Note: * indicates the level of significance at 5%

Table 5: Long & short run relationship between investor sentiment indices & market performance

Panel A: Cointegration (ARDL Model)					
Dependent Variable	Independent Variable	ARDL Lags	F statistic	Outcome	
BW Index	Sensex	1,1	16.100*	Cointegration	
BWY Index	Sensex	1,1	15.378*	Cointegration	
Sensex	Composite Index	1,1	6.3785*	Cointegration	
Aligned Investor Index	Sensex	1,1	30.330*	Cointegration	
Panel B: Short Run Dynamics (ECT Model)					
Error Correction Coefficient	BW Index	Sensex	BWY Index	Sensex	
	-0.7548 [0.1302]	-0.4207 [0.4158]	-0.7785 [0.1475]	-0.3345 [0.4197]	
	-5.7981*	-1.0119	-5.2769*	-0.7968	
Lead/Lag	Lagging	Leading	Lagging	Leading	
Error Correction Coefficient	Composite Index	Sensex	Aligned Investor Index	Sensex	
	-0.5758 [0.2213]	-0.7747 [0.2172]	-1.0856 [0.1515]	-0.6069 [0.3753]	
	-2.6016*	-3.5662*	-7.1629*	-1.6169	
Lead/Lag	Leading	Lagging	Lagging	Leading	
Panel C: Granger Causality					
Null Hypothesis	F statistic	P value	Null Hypothesis	F statistic	P value
BW Index doesn't granger cause Sensex	3.0314	0.0838	Composite Index doesn't granger cause Sensex	23.157	0.000*
Sensex doesn't granger cause BW Index	0.9392	0.3341	Sensex doesn't granger cause Composite Index	0.1041	0.7471
BWY Index doesn't granger cause Sensex	0.1273	0.7218	Aligned Investor Index doesn't granger cause Sensex	0.1049	0.7465
Sensex doesn't granger cause BWY Index	1.4865	0.2247	Sensex doesn't granger cause Aligned Investor Index	0.2677	0.6058

Notes: a) * indicates level of significance at 5%, values in parentheses, [] show t-values.

b) The lag structure is decided based on the minimum values of the Schwartz information Criterion

c) The relevant critical value bounds for f statistic are obtained from Table C1.iii (with an unrestricted intercept and no trend; with one regressor k=1) in Pesaran et al. (2001). They are 4.94 – 5.73 at 95% confidence interval

Table 6: Long and short run relationship between investor sentiment indices

Panel A: Cointegration Test						
Dependent Variable	Independent Variable	ARDL Lags	F statistic	Outcome		
BW Index	Composite Index	1,1	18.214*	Cointegration		
BWY Index	Composite Index	1,1	29.397*	Cointegration		
Aligned Investor Index	Sensex	1,2	28.476*	Cointegration		
Panel B: Short Run Dynamics (ECT Model)						
	BW Index	Composite Index	BWY Index	Composite Index	Aligned Investor Index	Composite Index
Error Correction Coefficient	-0.8026	-0.5834	-0.9318	-0.6481	-0.8938	-0.7411
	[0.1301]	[0.2037]	[0.1226]	[0.2004]	[0.1423]	[0.1857]
	-6.167*	-2.863*	-7.598*	-3.234*	-6.277*	-3.988*
Lead/Lag	Lagging	Leading	Lagging	Leading	Lagging	Leading

Notes: a) * indicates level of significance at 5%, values in parentheses, [] show t-values.

b) The lag structure is decided based on the minimum values of the Schwartz information Criterion for all cases.

c) The relevant critical value bounds for f statistic are obtained from Table C1.iii (with an unrestricted intercept and no trend; with one regressor k=1) in Pesaran et al. (2001). They are 4.94 – 5.73 at 95% confidence interval

Table 7: Unadjusted returns for size value and momentum sorted portfolios

Panel A: Quintiles										
Portfolio	P1	P2	P3	P4	P5					
Size sorted	0.0133	0.0167	0.0193	0.0267	0.0409					
Value sorted	0.0186	0.0198	0.0221	0.0259	0.0303					
Momentum	0.0273	0.0176	0.0154	0.0175	0.0137					
Panel B: Deciles										
Portfolio	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
Size sorted	0.0282	0.0155	0.0176	0.0157	0.0196	0.0187	0.0255	0.0278	0.0354	0.0462
Value sorted	0.0266	0.0168	0.0197	0.0207	0.0226	0.0209	0.0251	0.0259	0.0274	0.0333
Momentum	0.0190	0.0263	0.0191	0.0162	0.0164	0.0141	0.0165	0.0181	0.0186	0.0090
Panel C: Vigintiles										
Portfolio	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
Size sorted	0.0263	0.0293	0.0124	0.0188	0.0174	0.0178	0.0157	0.0156	0.0180	0.0212
Value sorted	0.0248	0.0280	0.0169	0.0167	0.0180	0.0214	0.0216	0.0200	0.0234	0.0223
Momentum	0.0180	0.0137	0.0333	0.0195	0.0203	0.0180	0.0146	0.0178	0.0174	0.0154
Portfolio	P11	P12	P13	P14	P15	P16	P17	P18	P19	P20
Size sorted	0.0171	0.0204	0.0247	0.0264	0.0263	0.0293	0.0332	0.0376	0.0438	0.0466
Value sorted	0.0232	0.0186	0.0233	0.0270	0.0265	0.0247	0.0221	0.0327	0.0267	0.0400
Momentum	0.0109	0.0172	0.0194	0.0134	0.0202	0.0160	0.0175	0.0200	0.0157	0.0010

Table 8: CAPM based results for sample portfolios

Panel A: Quintiles											
Portfolio		P1	P2	P3	P4	P5					
Size	α	0.0015 [1.19]	0.0039 [1.62]	0.0064 [2.08]*	0.0139 [3.68]*	0.0281 [5.19]*					
	β	1.0537 [66.6]*	1.1342 [36.9]*	1.1479 [29.6]*	1.1387 [23.9]*	1.1387 [16.7]*					
	Adjusted R ²	0.9675	0.9016	0.8549	0.7922	0.6517					
Value	α	0.0079 [2.69]*	0.0075 [3.23]*	0.0098 [3.49]*	0.0127 [3.66]*	0.0160 [3.38]*					
	β	0.9565 [25.9]*	1.0907 [37.2]*	1.0985 [31.1]*	1.1741 [26.9]*	1.28 [21.4]*					
	Adjusted R ²	0.8180	0.9025	0.8668	0.8295	0.7539					
Momentum	α	0.0171 [3.26]*	0.0075 [1.85]*	0.0047 [1.25]	0.0065 [1.54]	0.0013 [0.24]					
	β	0.9158 [13.8]*	0.8942 [17.5]*	0.9544 [20.1]*	0.9831 [18.6]*	1.1048 [15.3]*					
	Adjusted R ²	0.5629	0.6724	0.7317	0.6975	0.6105					
Panel B: Deciles											
	Portfolio	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
Size	α	0.0164 [4.63]*	0.0035 [1.74]	0.0050 [2.10]*	0.0029 [0.97]	0.0067 [2.14]*	0.0058 [1.69]	0.0127 [3.16]*	0.0149 [3.66]*	0.0219 [4.69]*	0.0342 [4.54]*
	β	1.0500 [23.6]*	1.0708 [41.9]*	1.1195 [37.3]*	1.1458 [30.4]*	1.1459 [28.9]*	1.1513 [26.6]*	1.1343 [22.4]*	1.1485 [22.4]*	1.2038 [20.5]*	1.0668 [11.2]*
	Adjusted R ²	0.7887	0.9225	0.9034	0.8611	0.8483	0.8259	0.7699	0.7703	0.7374	0.4568
Value	α	0.0138 [3.78]*	0.0062 [2.69]*	0.0072 [2.68]*	0.0088 [3.33]*	0.0105 [3.97]*	0.0083 [2.30]*	0.0122 [3.43]*	0.0124 [3.11]*	0.0136 [3.03]*	0.0186 [3.29]*
	β	1.1391 [24.8]*	0.9434 [32.5]*	1.1165 [33.1]*	1.0627 [32.2]*	1.0688 [31.8]*	1.1283 [24.9]*	1.1514 [25.8]*	1.1972 [23.8]*	1.2344 [21.9]*	1.3102 [18.5]*
	Adjusted R ²	0.8057	0.8766	0.8804	0.8742	0.8718	0.8069	0.8169	0.7911	0.7621	0.6953
Momentum	α	0.0073 [1.42]	0.0167 [2.70]*	0.0093 [2.14]*	0.0061 [1.45]	0.0058 [1.56]	0.0032 [0.79]	0.0057 [1.301]	0.0069 [1.50]	0.0061 [1.14]	-0.003 [-0.51]
	β	1.0378 [16.0]*	0.8578 [11.0]*	0.8774 [16.1]*	0.9069 [17.2]*	0.9418 [20.1]*	0.9644 [18.6]*	0.9555 [18.1]*	0.9963 [17.2]*	1.1190 [16.71]*	1.1039 [13.19]*
	Adjusted R ²	0.6325	0.4471	0.6353	0.6658	0.7321	0.6991	0.6879	0.6645	0.6536	0.5372
Panel C: Vigintiles											
Size	Portfolio	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
	α	0.0134 [4.18]*	0.0179 [2.99]*	0.0012 [0.53]	0.0059 [2.13]*	0.0045 [1.67]	0.0056 [1.77]	0.0023 [0.59]	0.0033 [1.12]	0.0061 [1.76]	0.0075 [2.07]*
	β	1.1531 [28.6]*	1.0068 [13.3]*	1.0002 [35.1]*	1.1413 [32.4]*	1.1485 [34.1]*	1.0908 [27.5]*	1.1973 [24.3]*	1.0982 [29.5]*	1.0647 [24.7]*	1.2252 [26.8]*
	Adjusted R ²	0.8465	0.5427	0.8917	0.8761	0.8866	0.8351	0.7986	0.8541	0.8037	0.8285
		P11	P12	P13	P14	P15	P16	P17	P18	P19	P20
	α	0.0034 [0.87]	0.0083 [2.25]*	0.0121 [2.77]*	0.0136 [3.01]*	0.0133 [3.03]*	0.0165 [3.61]*	0.0195 [3.99]*	.0242 [4.57]*	0.0302 [5.06]*	0.0363 [2.88]*
	β	1.2212 [24.7]*	1.0775 [23.3]*	1.1222 [20.3]*	1.1467 [20.3]*	1.1571 [20.9]*	1.1415 [19.8]*	1.2143 [19.7]*	1.1927 [17.9]*	1.2055 [16.1]*	0.9183 [5.79]*

Value	Adjusted R ²	0.8031	0.7843	0.7345	0.7331	0.7465	0.7241	0.7213	0.6817	0.6321	0.1795
	Portfolio	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
	α	0.0118 [3.76]*	0.0148 [3.75]*	0.0063 [2.25]*	0.0061 [2.21]*	0.0058 [1.87]	0.0085 [2.66]*	0.0098 [3.09]*	0.0079 [2.43]*	0.0109 [3.48]*	0.0107 [3.06]*
	β	1.1526 [29.1]*	1.1691 [23.5]*	0.9401 [26.4]*	0.9461 [27.4]*	1.0882 [27.9]*	1.1433 [28.3]*	1.0549 [26.4]*	1.0709 [25.9]*	1.1037 [27.8]*	1.0331 [23.4]*
	Adjusted R ²	0.8494	0.7872	0.8239	0.8344	0.8401	0.8434	0.8246	0.8187	0.8386	0.7856
	Portfolio	P11	P12	P13	P14	P15	P16	P17	P18	P19	P20
	α	0.011 [2.66]*	0.0058 [1.39]	.0102 [2.74]*	0.0144 [3.47]*	0.0132 [2.85]*	0.0111 [2.41]*	0.0087 [2.02]*	0.0184 [3.26]*	0.0125 [2.28]*	0.0247 [3.39]*
	β	1.1193 [22.2]*	1.1359 [21.6]*	1.1714 [25.1]*	1.1275 [21.6]*	1.1832 [20.2]*	1.2159 [21.1]*	1.1897 [21.8]*	1.2756 [17.9]*	1.2645 [18.3]*	1.3611 [14.8]*
	Adjusted R ²	0.7673	0.7579	0.8077	0.7578	0.7327	0.7484	0.7618	0.6823	0.6925	0.5956
	Portfolio	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
	α	0.0066 [1.58]	0.0022 [0.39]	0.0243 [2.52]*	0.0092 [1.85]	0.0105 [2.26]*	0.0080 [1.77]	0.0044 [0.97]	0.0077 [1.71]	0.0071 [1.79]	0.0046 [1.11]
	β	1.0137 [19.2]*	1.0215 [14.5]*	0.8012 [6.62]*	0.9152 [14.6]*	0.8671 [14.7]*	0.8868 [15.5]*	0.9111 [16.1]*	0.9017 [15.9]*	0.9204 [18.6]*	0.9634 [18.6]*
	Adjusted R ²	0.7119	0.5860	0.2235	0.5893	0.5928	0.6164	0.6314	0.6301	0.6981	0.6995
	Portfolio	P11	P12	P13	P14	P15	P16	P17	P18	P19	P20
	α	0.0002 [0.04]	0.0063 [1.29]	0.0088 [1.96]*	0.0026 [0.57]	0.0088 [1.76]	0.0051 [1.08]	0.0045 [0.82]	0.0078 [1.36]	0.0035 [0.57]	-0.012 [-1.33]
	β	0.9566 [19.1]*	0.9726 [15.7]*	0.9483 [16.8]*	0.9617 [16.4]*	1.0116 [15.9]*	0.9814 [16.9]*	1.1549 [16.7]*	1.0837 [14.9]*	1.0893 [14.1]*	1.1317 [10.2]*
	Adjusted R ²	0.7087	0.6248	0.6539	0.6427	0.6311	0.6559	0.6498	0.5991	0.5675	0.4129

Note: * indicates the 5% level of significance, coefficient values and [] show t-values. Coefficient β refers to excess markets returns
OLS results based on Newey West estimation of least squares which gives hetroskedasticity and autocorrelation corrected SE

Table 9: FF3F based results for sample portfolios whose returns are not fully explained by CAPM

Panel A: Quintiles						
Size Sorted	Portfolio	P3	P4	P5		
	α	-0.001 [-0.07]	0.0023 [0.87]	0.0059 [2.43]*		
	β	1.0754 [15.4]*	1.0396 [20.7]*	1.0683 [26.8]*		
	γ	0.4839 [7.12]*	0.8603 [10.4]*	1.6528 [12.6]*		
	δ	0.4634 [10.2]*	0.6554 [8.85]*	0.5819 [3.96]*		
	Adjusted R ²	0.9063	0.9171	0.9226		
Value Sorted	Portfolio	P1	P2	P3	P4	P5
	α	0.0019 [1.11]	0.0027 [1.23]	0.0029 [1.09]	0.0035 [1.61]	0.0028 [1.42]

	β	1.0207 [18.6]*	1.0957 [20.2]*	1.028 [33.1]*	1.0491 [21.4]*	1.0847 [24.8]*					
	γ	0.4599 [3.70]*	0.3658 [4.99]*	0.5078 [6.90]*	0.6711 [11.3]*	0.9561 [14.6]*					
	δ	-0.347 [-2.96]*	0.0136 [0.22]	0.4519 [6.74]*	0.7807 [13.3]*	1.1839 [18.4]*					
	Adjusted R ²	0.8903	0.9209	0.9252	0.9474	0.9552					
Momentum Sorted	Portfolio	P1	P2								
	α	0.0061 [1.56]	0.0019 [0.62]								
	β	0.9147 [8.23]*	0.8547 [8.86]*								
	γ	0.8261 [4.16]*	0.4112 [3.38]*								
	δ	0.0987 [0.46]	0.2685 [2.03]*								
	Adjusted R ²	0.6447	0.7038								
Panel B: Deciles											
Size Sorted	Portfolio	P1	P3	P5	P7	P8	P9	P10			
	α	0.0040 [1.66]	0.0037 [1.79]	0.0019 [0.72]	0.0021 [0.69]	0.0023 [0.84]	0.0058 [2.43]*	0.0059 [1.21]			
	β	1.0509 [32.1]*	1.0617 [26.1]*	1.0801 [15.4]*	1.0238 [17.5]*	1.0605 [20.8]*	1.0815 [23.5]*	1.0499 [19.9]*			
	γ	0.9279 [7.08]*	0.0908 [1.55]	0.3530 [4.63]*	0.7757 [8.04]*	0.9356 [10.1]*	1.1830 [14.4]*	2.1286 [8.03]*			
	δ	0.0983 [0.74]	0.3336 [6.50]*	0.4109 [7.15]*	0.7097 [9.69]*	0.6012 [6.61]*	0.8221 [10.3]*	0.3333 [1.21]			
	Adjusted R ²	0.9035	0.9217	0.8819	0.8874	0.8951	0.9216	0.7981			
Value Sorted	Portfolio	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
	α	0.0016 [0.69]	0.0022 [1.18]	0.0026 [1.16]	0.0033 [1.24]	0.0049 [1.45]	0.0006 [0.22]	0.0041 [1.82]	0.0021 [0.71]	0.0042 [1.75]	0.0019 [0.64]
	β	1.0977 [20.4]*	0.9595 [20.9]*	1.1348 [16.5]*	1.0468 [25.5]*	1.0523 [39.9]*	1.0155 [20.5]*	1.0293 [21.3]*	1.0691 [19.1]*	1.0624 [19.6]*	1.0994 [23.6]*
	γ	0.9081 [6.19]*	0.3034 [4.14]*	0.3444 [4.32]*	0.4035 [4.89]*	0.4259 [5.66]*	0.5576 [5.79]*	0.5838 [6.24]*	0.7582 [9.57]*	0.6737 [8.94]*	1.2146 [13.1]*
	δ	0.3353 [2.32]*	-0.056 [-1.07]	-0.065 [-1.09]	0.1350 [1.74]	0.1403 [1.87]	0.6987 [8.76]*	0.7534 [8.32]*	0.8072 [9.08]*	1.0454 [18.3]*	1.3249 [11.7]*
	Adjusted R ²	0.9064	0.8948	0.8972	0.8974	0.8974	0.9011	0.9219	0.9142	0.9125	0.9358
Momentum Sorted	Portfolio	P2	P3								
	α	0.0033 [0.77]	0.0029 [0.86]								
	β	0.8638 [7.33]*	0.8261 [8.19]*								
	γ	1.0089 [2.84]*	0.4616 [3.49]*								
	δ	0.0783 [0.19]	0.3406 [1.99]*								

	Adjusted R ²	0.5591	0.6788								
Panel C: Vigintiles											
Size Sorted	Portfolio	P1	P2	P4	P10	P12	P13	P14	P15	P16	P17
	α	0.0027 [1.53]	0.0019 [0.41]	0.0061 [2.01]*	0.0023 [0.84]	-0.0005 [-0.17]	0.0018 [0.62]	0.0027 [0.64]	.0019 [0.57]	0.0025 [0.83]	0.0047 [1.68]
	β	1.0673 [22.8]*	1.0558 [17.1]*	1.0847 [26.7]*	1.1713 [11.8]*	0.9881 [18.3]*	1.0035 [15.7]*	1.0451 [16.8]*	1.0533 [14.6]*	1.0705 [25.7]*	1.0964 [15.8]*
	γ	0.7879 [12.6]*	1.2142 [3.91]*	-0.019 [-0.21]	0.3841 [4.25]*	0.6468 [9.79]*	0.7579 [7.24]*	0.7945 [6.32]*	0.8348 [7.91]*	1.0386 [9.97]*	1.0989 [11.8]*
	δ	0.5716 [11.3]*	-0.141 [-0.46]	0.3168 [4.27]*	0.3473 [5.09]*	0.5772 [10.7]*	0.7546 [10.7]*	0.6626 [5.71]*	0.6788 [5.97]*	0.5171 [5.14]*	0.7878 [6.46]*
	Adjusted R ²	0.9518	0.7018	0.8915	0.8531	0.8735	0.8538	0.8371	0.8591	0.8553	0.8767
	Portfolio	P18	P19	P20							
	α	0.0071 [2.31]*	0.0104 [3.31]*	-0.001 [-0.09]							
	β	1.0664 [23.4]*	1.0906 [27.2]*	0.9945 [9.05]*							
	γ	1.2668 [10.2]*	1.4722 [9.99]*	2.8022 [4.61]*							
	δ	0.8542 [11.3]*	0.8126 [6.76]*	-0.117 [-0.19]							
	Adjusted R ²	0.8759	0.8442	0.5123							
Value Sorted	Portfolio	P1	P2	P3	P4	P6	P7	P8	P9	P10	P11
	α	0.0024 [1.43]	0.0033 [0.97]	0.0029 [1.17]	0.0014 [0.53]	0.0039 [1.43]	0.0045 [1.57]	0.0023 [0.61]	0.0057 [1.31]	0.0042 [1.04]	0.0023 [0.61]
	β	1.0514 [25.4]*	1.0901 [20.1]*	0.9703 [19.1]*	0.9448 [19.2]*	1.1862 [19.9]*	1.0411 [26.1]*	1.0500 [17.7]*	1.0973 [22.3]*	1.002 [28.9]*	1.0244 [22.8]*
	γ	0.6919 [9.79]*	0.8545 [9.39]*	0.2606 [2.92]*	0.3464 [4.42]*	0.3549 [3.24]*	0.3929 [4.01]*	0.4189 [3.89]*	0.3941 [3.19]*	0.4889 [4.13]*	0.6092 [4.83]*
	δ	0.6485 [10.1]*	0.5414 [6.42]*	-0.141 [-1.85]	0.0275 [0.54]	-0.202 [-2.91]*	0.1215 [1.55]	0.1649 [1.28]	0.0802 [0.69]	0.2304 [2.03]*	0.6035 [7.68]*
	Adjusted R ²	0.9523	0.8886	0.8412	0.8536	0.8673	0.8446	0.8418	0.8568	0.8206	0.8471
	Portfolio	P13	P14	P15	P16	P17	P18	P19	P20		
	α	0.0029 [1.03]	.0053 [1.76]	0.0021 [0.59]	0.0016 [0.49]	-0.001 [-0.01]	0.0084 [2.51]*	-0.001 [-0.08]	0.0037 [0.67]		
	β	1.0672 [19.6]*	0.9885 [18.5]*	1.0642 [14.1]*	1.0808 [19.1]*	1.0586 [23.7]*	1.0631 [14.4]*	1.0665 [18.9]*	1.1354 [20.9]*		
	γ	0.5231 [4.17]*	0.6588 [8.18]*	0.8151 [6.68]*	0.6853 [6.32]*	0.6379 [5.84]*	0.7191 [6.45]*	0.9188 [8.06]*	1.5362 [12.5]*		
	δ	0.6464 [7.15]*	0.8576 [7.71]*	0.7623 [7.95]*	0.8358 [8.82]*	0.8111 [13.1]*	1.2795 [11.1]*	1.2197 [11.3]*	1.4447 [9.31]*		
	Adjusted R ²	0.8826	0.8887	0.8482	0.8579	0.8694	0.8621	0.8835	0.8551		
Momentum Sorted	Portfolio	P3	P5	P13							
	α	0.0051 [0.89]	0.0045 [1.08]	0.0046 [1.39]							

β	0.8695 [5.61]*	0.8213 [7.89]*	0.8827 [7.04]*
γ	1.4499 [2.16]*	0.3077 [2.91]*	0.3071 [2.05]*
δ	-0.224 [-0.31]	0.3076 [1.60]	0.4045 [2.46]*
Adjusted R ²	0.3856	0.6284	0.6853

Note: * indicates the 5% level of significance, coefficient values and [] show t-values. Coefficient β refers to excess markets returns, γ refers to SMB factor, δ refers to LMH factor. Multicollinearity has been observed if VIF greater than 10 and auxiliary regression has been run between correlated variables to transform the variables so that correlations between independent variables do not exceed 0.3 OLS results based on Newey West estimation of least squares which gives heteroskedasticity and autocorrelation corrected SE

Table 10: Augmented FF3F models for sample portfolios whose returns are not fully explained by FF3F models

Panel A: Carhart Model									
Group	Characteristic	Portfolio	α	β	γ	δ	λ	Adjusted R ²	
Quintile	Size Sorted	P5	0.0055 [2.25]*	1.0737 [25.1]*	1.6369 [14.7]*	0.5961 [4.58]*	0.0436 [0.63]	0.9227	
Decile	Size Sorted	P9	0.0065 [2.39]*	1.0789 [24.6]*	1.1893 [14.6]*	0.8143 [9.74]*	-0.0626 [-0.83]	0.9216	
Vigintile	Size Sorted	P4	0.0065 [2.23]*	1.0798 [24.1]*	-0.005 [-0.05]	0.3039 [4.31]*	-0.039 [-0.76]	0.8915	
Vigintile	Size Sorted	P18	0.0082 [2.51]*	1.0539 [27.3]*	1.3037 [10.9]*	0.8213 [10.8]*	-0.1013 [-1.65]	0.8784	
Vigintile	Size Sorted	P19	0.0106 [3.33]*	1.0875 [24.6]*	1.4813 [10.6]*	0.8044 [7.03]*	-0.025 [-0.36]	0.8433	
Vigintile	Value Sorted	P18	0.0096 [2.75]*	1.0493 [14.9]*	0.7596 [5.92]*	1.2434 [10.2]*	-0.111 [-1.42]	0.8646	
Panel B: FF 5 Factor Model									
Group	Characteristic	Portfolio	α	β	γ	δ	ν	φ	Adjusted R ²
Quintile	Size Sorted	P5	0.0058 [2.13]*	1.0701 [28.1]*	1.6635 [12.9]*	0.5789 [4.05]*	0.0369 [0.28]	-0.021 [-0.22]	0.9216
Decile	Size Sorted	P9	0.0042 [1.65]	1.0913 [24.3]*	1.3058 [12.6]*	0.8159 [11.7]*	0.1981 [2.75]*	-0.223 [-2.41]*	0.9274
Vigintile	Size Sorted	P4	0.0057 [1.94]	1.0963 [24.4]*	-0.141 [-0.95]	0.3214 [3.09]*	0.1365 [1.37]	-0.026 [-0.32]	0.8923
Vigintile	Size Sorted	P18	0.0051 [1.52]	1.1179 [24.3]*	1.1191 [6.97]*	1.0132 [9.56]*	0.2744 [3.37]*	-0.223 [-2.14]*	0.8837
Vigintile	Size Sorted	P19	0.0107 [3.54]*	1.0941 [22.3]*	1.0238 [4.28]*	0.6682 [4.77]*	0.3866 [2.31]*	0.1075 [0.99]	0.8558
Vigintile	Value Sorted	P18	0.0111 [2.85]*	0.9946 [11.4]*	0.7771 [4.14]*	1.0188 [5.46]*	-0.249 [-1.63]	0.3361 [2.01]*	0.8713
Panel C: Modified FF 5 Factor Model									
Group	Characteristic	Portfolio	α	β	γ	δ	ν	η	Adjusted R ²
Quintile	Size Sorted	P5	0.0054 [2.34]*	1.0825 [25.7]*	1.6835 [13.2]*	0.5767 [4.80]*	0.0063 [0.05]	-0.248 [-2.23]*	0.9293

Decile	Size Sorted	P9	0.0059 [2.37]*	1.0895 [24.1]*	1.1757 [13.9]*	0.7997 [11.1]*	0.1756 [2.14]*	0.027 [0.41]	0.9234	
-										
Vigintile	Size Sorted	P4	0.0061 [1.98]*	1.0906 [25.7]*	0.0258 [-0.28]	0.2995 [3.71]*	0.1367 [1.35]	0.0277 [0.46]	0.8924	
Vigintile	Size Sorted	P18	0.0072 [2.24]*	1.0771 [22.8]*	1.2541 [10.1]*	0.8219 [13.1]*	0.2554 [2.98]*	0.0573 [0.62]	0.8802	
Vigintile	Size Sorted	P19	0.0101 [3.48]*	1.1207 [26.1]*	1.4846 [10.6]*	0.7601 [7.86]*	0.3796 [2.31]*	-0.1689 [-1.31]	0.8581	
Vigintile	Value Sorted	P18	0.0083 [2.34]*	1.0529 [14.7]*	0.7271 [6.77]*	1.3069 [11.2]*	-0.2132 [-1.48]	-0.0261 [-0.34]	0.8637	
Panel D1: Investor Sentiment Factor Augmented FF 5 Factor Model										
Group	Characteristic	Portfolio	α	β	γ	δ	ν	ϕ	dSentiment	Adjusted R ²
Quintile	Size Sorted	P5	0.0033 [1.09]	1.0565 [27.3]*	1.6612 [12.9]*	0.5811 [4.06]*	0.0422 [0.32]	-0.0299 [-0.31]	0.0068 [1.29]	0.9211
Vigintile	Size Sorted	P19	0.0064 [1.47]	1.0888 [22.8]*	1.403 [10.0]*	0.7522 [7.45]*	0.3944 [2.34]*	0.0929 [0.83]	0.0128 [1.67]	0.8569
Panel D2: Investor Sentiment Factor Augmented FF 3 Factor Model										
Group	Characteristic	Portfolio	α	β	γ	dSentiment	Adjusted R ²			
Vigintile	Value Sorted	P18	0.0083 [1.87]	1.0599 [13.8]*	0.7127 [6.31]*	1.2842 [10.9]*	0.0007 [0.11]			

Note: * indicates the 5% level of significance, coefficient values and [] show t-values. Coefficient β refers to excess markets returns, γ refers to SMB factor, δ refers to LMH factor, λ refers to momentum factor, ν refers to investment factor, ϕ refers to profitability factor (ROE), η refers to cashflow variability and dSentiment refers to investor sentiment factor.

In modified FF 5 factor model, the profitability factor is replaced by cashflow variability factor. The model is inspired by the work of (Walkshausl, 2013) who use both profitability and cashflow variability as alternative measures of firm quality.

Multicollinearity has been observed if VIF greater than 10 and auxiliary regression has been run to transform the variables so that correlations between independent variables do not exceed 0.3

OLS results based on Newey West estimation of least squares which gives heteroskedasticity and autocorrelation corrected Std Error.

Figure 1: Diagram of average of cumulative returns over number of months for quintile size sorted P5 portfolio

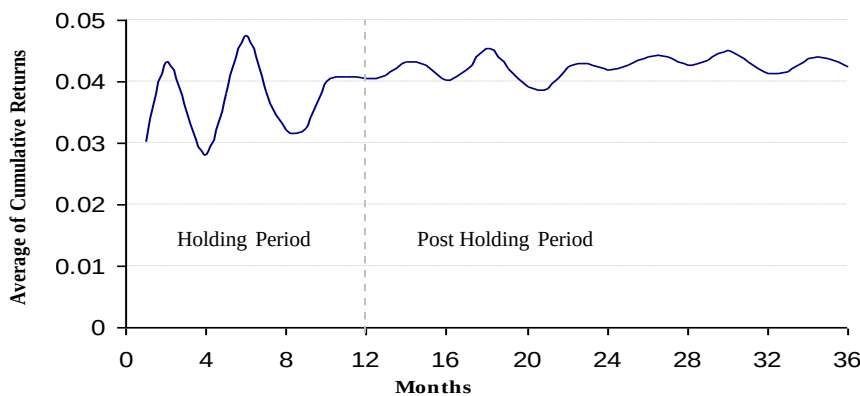


Figure 2: Diagram of average of cumulative returns over number of months for vigintile size sorted P19 portfolio

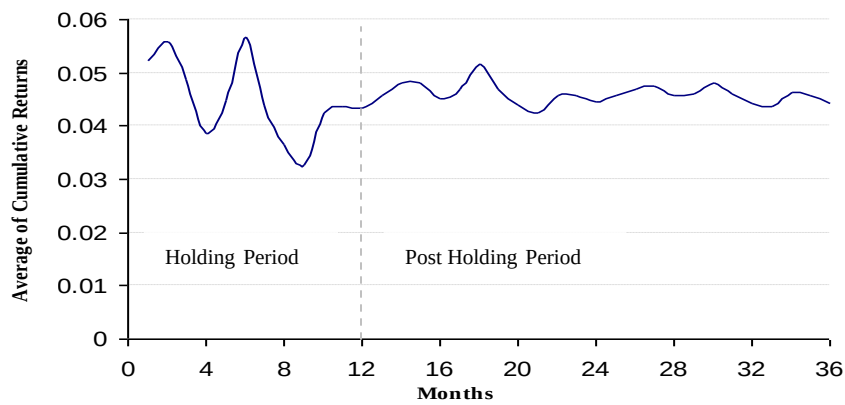


Figure 3: Diagram of average of cumulative returns over number of months for vigintile value sorted P18 portfolio

